Product Data Exploration Using Excel

Submitted by: Mookambika **Course:** Data Analytics

Overview

This project leverages Microsoft Excel to explore and analyse product data through a range of built-in formulas and functions, covering:

- **Calculations** — numerical operations on product metrics
- **Logical operations** — condition-based decision making
- **Conditional analysis** — data filtering and trend identification
- **Text formatting** — structuring and presenting data clearly

The goal is to derive meaningful insights from raw product data using Excel as the primary analytical tool.

Dataset Description

Column Name	Data Description	Analytical Purpose
Product ID	A unique identifier assigned to each product record	Used as a primary key to distinguish individual records
Day	Numeric day value extracted from the Product ID	Enables time-based analysis at the daily level
Country Code	Abbreviated code representing the product's country	Supports geographic segmentation and regional comparison
Month	Numeric month value extracted from the Product ID	Facilitates monthly trend analysis and time series exploration
Product Name	The name or title of the product	Used for product-level filtering and categorisation

Brand Name	The brand or manufacturer of the product	Enables brand-wise performance comparison
Price ($)	The selling price of the product in US dollars	Key metric for pricing analysis and statistical computation
Quantity	The number of units of each product	Used to measure demand, volume, and inventory patterns
Category	The product type or segment it belongs to	Supports category-wise grouping and comparative analysis
Price Range	Classification of products as Low, Medium, or High price	Enables conditional analysis and market segmentation

Data Cleaning & Transformation

The raw dataset was cleaned and transformed using Excel's built-in text functions to extract meaningful variables from the Product ID field.

Transformation	**Function**	**Output**
Extracted **Day** from Product ID	=LEFT(A2,2)	28
Extracted **Country Code** from Product ID	=RIGHT(A3,2)	US
Extracted **Month** from Product ID	=MID(A2,4,3)	JAN
Classified products by price level	=IF(G2>=500,"High Price", "Standard Price")	High Price / Standard Price

Visualization and Insights Arrived with screenshot of visualization Report/Dashboard screenshot:

Analysis and Insights

1. Descriptive Analysis (What happened?)

Summarises the overall distribution and key statistics of the product data.

Metric	Function	Result
Total Price of All Products	=SUM(G2:G35)	$ 10,100
Total Number of Products	=COUNTA(E2:E35)	34
Average Product Price	=AVERAGE(G2:G35)	$ 297
Minimum Price	=MIN(G2:G35)	$ 30
Maximum Price	=MAX(G2:G35)	$ 1,000
Total Price — Electronics	=SUMIF(I2:I35,"Electronics",G2:G35)	$ 8,050
Products Priced Below $100	=COUNTIF(G2:G35,"<100")	11

2. Diagnostic Analysis (Why did it happen?)

Identifies patterns and reasons behind the observed data trends.

- **Electronics leads revenue** at $8,050 — driven by high unit prices across brands like Dell, HP, and Samsung.
- **11 products fall below $100** — indicating a strong budget product segment within the dataset.
- **Average price ($297) vs Maximum ($1,000)** — confirms pricing is skewed by premium items such as Laptops and Cameras.
- **IF() classification** reveals most products fall under Standard Price, with only select items marked High Price.

3. Predictive Analysis (What might happen?)

Forecasts likely trends based on existing data patterns.

- Electronics category is projected to **remain the top revenue contributor** given its consistent high pricing.

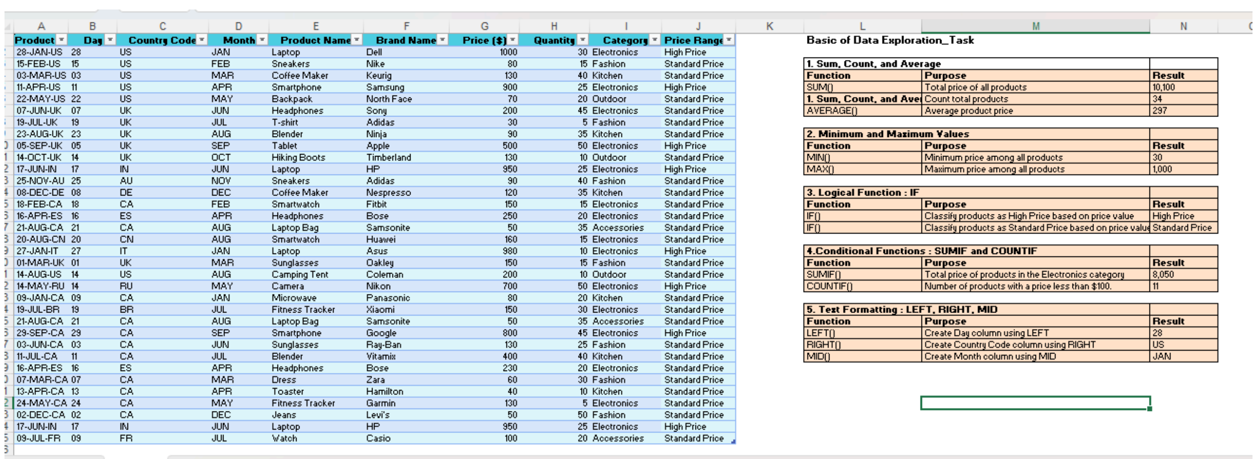
- Budget-segment products (below $100) suggest **high sales volume potential** through quantity-driven demand.
- Monthly extraction (JAN–DEC) indicates **seasonal purchasing patterns** that can guide demand forecasting.
- Regional patterns via Country Code point to **market-specific product preferences** across US, UK, CA, and IN.

4. Prescriptive Analysis (What should be done?)

Recommends data-driven actions to optimise product performance.

- **Prioritise Electronics** in procurement and marketing — highest revenue-generating category.
- **Bundle low-priced products** to increase average order value and improve overall sales margins.
- **Leverage monthly trends** to plan seasonal promotions and manage inventory effectively.
- **Use country-wise segmentation** to run targeted regional campaigns for high-performing markets.
- **Apply discount strategies** on High Price products to improve conversion and sales volume.

DashBoard Screen-short



Product	Day	Country Code	Month	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range
28-JAN-US	28	US	JAN	Laptop	Dell	1000	30	Electronics	High Price
15-FEB-US	15	US	FEB	Sneakers	Nike	80	15	Fashion	Standard Price
03-MAR-US	03	US	MAR	Coffee Maker	Keurig	130	40	Kitchen	Standard Price
11-APR-US	11	US	APR	Smartphone	Samsung	900	25	Electronics	High Price
22-MAY-US	22	US	MAY	Backpack	North Face	70	20	Outdoor	Standard Price
07-JUN-UK	07	UK	JUN	Headphones	Sony	200	45	Electronics	Standard Price
19-JUL-UK	19	UK	JUL	T-shirt	Adidas	30	5	Fashion	Standard Price
23-AUG-UK	23	UK	AUG	Blender	Ninja	90	35	Kitchen	Standard Price
05-SEP-UK	05	UK	SEP	Tablet	Apple	500	50	Electronics	High Price
14-OCT-UK	14	UK	OCT	Hiking Boots	Timberland	130	10	Outdoor	Standard Price
17-JUN-IN	17	IN	JUN	Laptop	HP	950	25	Electronics	High Price
25-NOV-AU	25	AU	NOV	Sneakers	Adidas	90	40	Fashion	Standard Price
08-DEC-DE	08	DE	DEC	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price
18-FEB-CA	18	CA	FEB	Smartwatch	Fitbit	150	15	Electronics	Standard Price
16-APR-ES	16	ES	APR	Headphones	Bose	250	20	Electronics	Standard Price
21-AUG-CA	21	CA	AUG	Laptop Bag	Samsonite	50	35	Accessories	Standard Price
20-AUG-CN	20	CN	AUG	Smartwatch	Huawei	160	15	Electronics	Standard Price
27-JAN-IT	27	IT	JAN	Laptop	Asus	980	10	Electronics	High Price
01-MAR-UK	01	UK	MAR	Sunglasses	Oakley	150	15	Fashion	Standard Price
14-AUG-US	14	US	AUG	Camping Tent	Coleman	200	10	Outdoor	Standard Price
14-MAY-RU	14	RU	MAY	Camera	Nikon	700	50	Electronics	High Price
09-JAN-CA	09	CA	JAN	Microwave	Panasonic	80	20	Kitchen	Standard Price
19-JUL-BR	19	BR	JUL	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price
21-AUG-CA	21	CA	AUG	Laptop Bag	Samsonite	50	35	Accessories	Standard Price
29-SEP-CA	29	CA	SEP	Smartphone	Google	800	45	Electronics	High Price
03-JUN-CA	03	CA	JUN	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price
11-JUL-CA	11	CA	JUL	Blender	Vitamix	400	40	Kitchen	Standard Price
16-APR-ES	16	ES	APR	Headphones	Bose	230	20	Electronics	Standard Price
07-MAR-CA	07	CA	MAR	Dress	Zara	60	30	Fashion	Standard Price
13-APR-CA	13	CA	APR	Toaster	Hamilton	40	10	Kitchen	Standard Price
24-MAY-CA	24	CA	MAY	Fitness Tracker	Garmin	130	5	Electronics	Standard Price
02-DEC-CA	02	CA	DEC	Jeans	Levi's	50	50	Fashion	Standard Price
17-JUN-IN	17	IN	JUN	Laptop	HP	950	25	Electronics	High Price
09-JUL-FR	09	FR	JUL	Watch	Casio	100	20	Accessories	Standard Price

Basic of Data Exploration_Task

1. Sum, Count, and Average		
Function	Purpose	Result
SUM()	Total price of all products	10,100
1. Sum, Count, and Ave	Count total products	34
AVERAGE()	Average product price	297

2. Minimum and Maximum Values		
Function	Purpose	Result
MIN()	Minimum price among all products	30
MAX()	Maximum price among all products	1,000

3. Logical Function : IF		
Function	Purpose	Result
IF()	Classify products as High Price based on price value	High Price
IF()	Classify products as Standard Price based on price valu	Standard Price

4.Conditional Functions : SUMIF and COUNTIF		
Function	Purpose	Result
SUMIF()	Total price of products in the Electronics category	8,050
COUNTIF()	Number of products with a price less than $100.	11

5. Text Formatting : LEFT, RIGHT, MID		
Function	Purpose	Result
LEFT()	Create Day column using LEFT	28
RIGHT()	Create Country Code column using RIGHT	US
MID()	Create Month column using MID	JAN